

Experiment 8: API Integration – Connect Applications to OpenAI, Google Gemini, or Hugging Face Inference APIs

Aim

To integrate a Python application with OpenAI, Google Gemini, and Hugging Face Inference APIs for generating AI-based responses.

Python Code:

OpenAI Code

```
from openai import OpenAI

client = OpenAI(
    api_key="sk-proj-FboLgW768fa2IYU2-KXDkmHy_cK9JgnW_P9A-A_3SStE-
Fm94OYnHlDpB6mo5qbWfba9c7LnGdT3BlbkFJoqosff_LYE2VyPCdbNGX_4p3ghTZhcd
Bt4Ff_Fh0tYqmNnhjB64OmSAx8KtGx7X9XtRDIN0cUA"
)

prompt = "Explain the importance of Artificial Intelligence in 100 words."

response = client.responses.create(
    model="gpt-5.4-mini",
    input=prompt
)

print("===== OPENAI API RESPONSE =====")
print(response.output_text)
```

Hugging Face Code

```
from huggingface_hub import InferenceClient

client = InferenceClient(
    api_key="hf_QGbcpxeaNlAsajmOVNfrPEAJolXiMegsr"
)

completion = client.chat.completions.create(
    model="Qwen/Qwen2.5-0.5B-Instruct:featherless-ai",
    messages=[
        {
            "role": "user",
            "content": "Explain the importance of Artificial Intelligence in 100 words."
        }
    ],
)

print(completion.choices[0].message.content)
```

Google Gemini Code

```
from google import genai
```

```
client = genai.Client(  
    api_key="AQ.Ab8RN6L3YAVP-nVNSVmBwU0enA921VTeWi-auPhO0ZLuDmwQxg"  
)
```

```
response = client.models.generate_content(  
    model="gemini-flash-latest",  
    contents="Explain the importance of Artificial Intelligence in 100 words."  
)
```

```
print(response.text)
```

Output

===== OPENAI API RESPONSE =====

Artificial Intelligence is important because it helps machines perform tasks that usually require human intelligence, such as learning, problem-solving, and decision-making. It improves efficiency by automating repetitive work, saves time, and reduces errors. AI is widely used in healthcare, education, finance, transportation, and customer service, making services faster and more accurate. It also helps analyze large amounts of data to find patterns and make better predictions. In daily life, AI powers virtual assistants, recommendations, and smart devices. As technology grows, AI will play an even bigger role in innovation, productivity, and solving complex global challenges.

===== Hugging Face Response =====

Artificial Intelligence (AI) is incredibly important because it can transform various sectors and create substantial benefits for society. In business, AI can optimize processes, automate tasks, and improve decision-making. In transportation, AI's autonomous systems can enhance safety and reduce accidents. In healthcare, AI can diagnose diseases, suggest treatment options, and even assist in medication management. Additionally, AI helps in analyzing vast amounts of data, providing insights that are difficult for humans to make definitively. Overall, AI positions businesses, industries, and society as more resilient and productive, depending on how they utilize its full capabilities.

===== Google Gemini Response =====

Artificial Intelligence is fundamentally transforming our world by automating complex tasks, boosting productivity, and driving unprecedented innovation across every industry. Its ability to analyze massive datasets instantly enables smarter decisions, accurate predictions, and personalized user experiences.

In healthcare, AI revolutionizes early disease diagnosis and accelerates novel drug discovery. Economically, it streamlines business operations and enhances overall efficiency.

Moreover, AI aids in solving critical global challenges, such as climate change and resource management.

By augmenting human capabilities, AI liberates us from routine labor, inspiring innovation and creativity. Ultimately, AI is an indispensable engine driving a smarter, safer, and more prosperous future.

Result

The Python application was successfully integrated with an AI API and generated the required response for the given prompt, demonstrating successful API integration and text generation.